SMAI Assignment 4 - Question 1 Report

Multi-Task CNN on Fashion-MNIST

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Base Configuration:

- Channels: [32, 64, 128]
- Dropout: 0.3
- Optimizer: Adam
- Learning Rate: 0.001
- Batch Size: 128
- Epochs: 5 (due to computational constraints)

Experimental Approach

Exploring the hyperparameters, I divided experiments into six groups:

Group	Parameters Varied	Purpose
G1 - Lambda Ratios	λ_1:λ_2 → [1:1, 2:1, 1:2]	Understand task trade-offs
G2 - Learning Rate	LR → [0.0001, 0.0005, 0.001, 0.005, 0.01]	Find optimal convergence speed
G3 - Dropout	dropout → [0.1, 0.3, 0.5, 0.7]	Balance regularization
G4 - Optimizers	[SGD, Adam, AdamW]	Compare optimization algorithms
G5 - Architecture	Depth & width variations	Test model capacity
G6 - Batch Size	[32, 64, 128, 256]	Analyze gradient stability

3. Results & Analysis

3.1 Lambda Ratio Exploration (Task Trade-off)

Theoretical Expectation:

- Λ_1 Inc → Model prioritizes classification → ↑ Accuracy, ↑ RMSE (worse regression)
- Λ_2 Inc → Model prioritizes regression → ↓ RMSE, ↓ Accuracy (worse classification)

My Results:

Configuration	λ_1:λ_2	Test Acc	Test RMSE
Balanced	1:1	0.8720	0.0351
Class Focus	2:1	**0.8813**	0.0318
Regression Focus	1:2	0.8741	**0.0399**

Analysis:

- **λ_1=2.0, λ_2=1.0**: Achieved highest accuracy (0.8813) as expected,
- **λ_1=1.0, λ_2=2.0**: Expected to achieve best RMSE, but got 0.0399 (worse than balanced)
    - **Reason**: Limited training (5 epochs) prevented the model from fully converging on the regression task
    - Regression requires more iterations to learn pixel-level patterns
- (1:1) provided reasonable performance on both tasks

3.2 Learning Rate Exploration

Theoretical Expectation:

- Very low LR: Slow but stable convergence
- Optimal LR: Fast convergence with stability
- Very high LR: Unstable training, potential divergence

My Results:

LR	Test Acc	Test RMSE
0.0001	0.8391	0.0326
0.0005	0.8758	0.0311
0.001	**0.8820**	**0.0298**
0.005	0.8766	0.3132 *
0.01	0.8592	0.3132 *

Analysis:

- **LR=0.001**: Optimal performance, balancing convergence speed and stability
- **LR=0.0001**: Under-performed due to insufficient convergence in 5 epochs
- **LR ≥ 0.005**: RMSE exploded to 0.3132, indicating **gradient instability** and overshooting

LR=0.001 is optimal for this architecture and dataset.

3.3 Dropout Rate Exploration

Theoretical Expectation:

- Low dropout: Better training fit, risk of overfitting
- High dropout: Strong regularization, may underfit

My Results:

Dropout	Test Acc	Test RMSE
0.1	**0.9020**	**0.0256**
0.3	0.8826	0.0332
0.5	0.8445	0.0480
0.7	0.7628	0.0637

Analysis:

- **Dropout=0.1 is the best performer**: Best accuracy (0.9020) AND best RMSE (0.0256)
- Clear degradation pattern as dropout increases, matching theory

- High dropout (0.7) causes severe underfitting, reducing accuracy to 76.28%

3.4 Optimizer Comparison

Theoretical Expectation:

- SGD: Slower, requires momentum
- Adam: Adaptive learning rates, faster convergence
- AdamW: Adam + weight decay regularization

My Results:

Optimizer	Test Acc	Test RMSE
SGD	0.7900	0.0512
Adam	**0.8855**	0.0351
AdamW	0.8757	0.0363

Analysis:

- **SGD significantly underperformed** (78% accuracy) due to:
  - No learning rate scheduling implemented
  - **Only 5 epochs insufficient** for SGD's slower convergence
- **Adam performed best** with adaptive learning rates
- AdamW showed slight regularization benefit

3.5 Architecture Exploration

My Results:

Architecture	Test Acc	Test RMSE
2-Block Shallow	0.8780	0.3132
3-Block Medium	0.8759	0.0340
3-Block Wide	**0.8913**	**0.0316**
3-Block Narrow	0.8633	0.0394

Analysis:

- **3-Block Wide [64, 128, 256]**: Best performance, confirming increased capacity helps
- The 2-Block architecture's RMSE explosion (0.3132) suggests **insufficient depth** for joint task learning
- Narrow architecture (16, 32, 64) underperformed due to **limited representational capacity**

3.6 Batch Size Exploration

My Results:

Batch Size	Test Acc	Test RMSE
32	**0.8853**	**0.0326**
64	0.8756	**0.0311**
128	0.8779	0.0346
256	0.8806	0.0425

Analysis:

- Smaller batches (32, 64) provide **noisy but effective gradients.**
- Larger batches (256) showed degraded performance, likely due to:
    - Reduced gradient updates per epoch (fewer iterations)
    - Less regularization effect from gradient noise

3.7 Combined Hyperparameter Search

After identifying optimal parameters from isolated experiments, I ran 5 strategic combinations:

Configuration	Test Acc	Test RMSE	Test MAE
Combined_Best_Overall	0.9132	0.0226	0.0163
Combined_Max_Classification	0.9124	0.0200	0.0155
Combined_Max_Regression	0.9116	**0.0195**	**0.0149**
Combined_Balanced_Fast	0.9157	0.0202	0.0157

Combined_Wide_Aggressive	**0.9166**	**0.0183**	**0.0140**

Best Configuration (Combined_Wide_Aggressive):

- λ_1=1.5, λ_2=1.0
- Channels: [64, 128, 256]
- Dropout: 0.1
- LR: 0.0005
- Batch: 64
- Optimizer: AdamW
- **Epochs: 10**

Combined search significantly outperformed isolated experiments:

- **Accuracy improved from 0.9020 → 0.9166** (+1.46%)
- **RMSE improved from 0.0256 → 0.0183** (-28.5%)

4. Feature Map Visualization & Interpretation

I generated feature maps for 3 test images across all 3 convolutional blocks, visualizing:

- **Grid view**: Combined 12-channel visualization per layer
- **Individual view**: Separate visualization of each channel

Layer-wise Analysis:

Block	Features Learned	Spatial Detail	Semantic Meaning	Contribution
Conv Block 1	Edges, intensity transitions, basic textures	**High**	Low	Detects object boundaries, distinguishes ink concentration at edges

Conv Block 2	Clothing textures, folds, regional shapes	**Medium**	Medium	Identifies fabric patterns, helps differentiate similar items (trouser vs jeans)
Conv Block 3	Abstract semantic shapes, overall structure	**Low**	**High**	Focuses on category-defining features, estimates total ink coverage

Observations:

1. **Encoder Block 1:**
    a. Strongly reacts to edges and contrast boundaries
    b. Preserves spatial structure (recognizable as clothing item)
    c. Critical for ink regression by identifying high-density pixel regions
2. **Encoder Block 2:**
    a. Background noise diminishes
    b. Texture patterns emerge (wrinkles, fabric type)
    c. Provides category-discriminative features
3. **Encoder Block 3:**
    a. Highly abstract, blocky representations
    b. Spatial details lost but semantic meaning retained
    c. Essential for final classification decision
    d. Helps regression by capturing global silhouette coverage

Task-Specific Insights:

- **Classification** benefits most from **Block 3** (semantic understanding)
- **Regression** benefits from **Blocks 1 & 2** (pixel-level intensity patterns)
- The shared backbone efficiently learns features useful for both tasks

5. Lambda Trade-off Analysis

Theory vs Reality:

Expected Behavior:

High λ_1 → Better Classification, Worse Regression
High λ_2 → Better Regression, Worse Classification

My Results:

λ_1:λ_2	Acc	RMSE	Matches Theory?
2:1	0.8813	0.0318	Partial (limited epochs)
1:2	0.8741	0.0399	Expected RMSE improvement
1:1	0.8720	0.0351	Balanced

Reasons for Mismatch:

1. **Insufficient Training (5 epochs)**:
    a. Classification converges faster (categorical patterns)
    b. Regression requires more epochs (continuous pixel-level learning)
2. **Random Seed Variation**:
    a. Different initialization can affect convergence trajectories
3. **Small Dataset Split**:
    a. Fashion-MNIST's simplicity may reduce trade-off visibility

6. Final Model Selection

Best Models:

Model	Configuration	Performance
best_accuracy_model	λ_1=1.5, λ_2=1.0, Wide arch, Dropout=0.1, AdamW, Batch=64	Acc: **0.9166**, RMSE: 0.0183
best_regression_model	λ_1=1.0, λ_2=2.0, Medium arch, Dropout=0.1, AdamW, Batch=32	Acc: 0.9116, RMSE: **0.0195**

Best Configuration:

```
{
   'lambda1': 1.5, 'lambda2': 1.0,
```

```
    'channels': [64, 128, 256],
    'dropout': 0.1,
    'lr': 0.0005,
    'batch_size': 64,
    'optimizer': 'AdamW',
    'epochs': 10
}
Accuracy: 91.66%, RMSE: 0.0183
```

Wandb Dashboard

All experiments logged at: https://wandb.ai/tempakash48-iiit-hyderabad/SMAI_A4_Q1_MultiTask_re_run?nw=nwusertempakash48